

CHACHOENGSAO AGROMET, Thailand

WMO: 484580

Lat: 13.5672N

Lon: 101.4544E

Elev: 70

StdP: 100.48

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 16.4	(c) 17.8	(d) 11.1	(e) 8.3	(f) 21.6	(g) 12.9	(h) 9.4	(i) 23.1	(j) 5.7	(k) 24.9	(l) 5.1	(m) 26.1	(n) 0.3	(o) N/A	(p) 0.295

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 9.5	(c) 35.9	(d) 26.2	(e) 35.0	(f) 25.8	(g) 34.2	(h) 25.6	(i) 27.6	(j) 32.4	(k) 27.2	(l) 31.9	(m) 27.0	(n) 31.5	(o) 1.8	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.5	(b) 22.1	(c) 29.1	(d) 26.2	(e) 21.7	(f) 28.8	(g) 25.9	(h) 21.4	(i) 28.5	(j) 87.8	(k) 32.7	(l) 86.4	(m) 32.0	(n) 85.3	(o) 31.6	(p) 29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 4.5	(b) 3.7	(c) 3.1	(e) 13.8	(f) 37.0	(g) 1.8	(h) 1.3	(i) 12.5	(j) 38.0	(k) 11.4	(l) 38.7	(m) 10.4	(n) 39.4	(o) 9.0	(p) 40.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	25.6	27.2	28.4	29.4	29.3	28.7	28.1	28.2	27.8	27.7	27.0	25.6
	DBStd	1.88	2.08	1.66	1.60	1.55	1.36	1.06	1.15	0.94	1.03	0.98	1.50	2.12
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6478	485	482	569	581	597	562	562	563	534	549	510	484
	CDD18.3	3436	227	249	311	331	339	312	303	305	284	291	260	226
	CDH23.3	34990	2104	2534	3328	3711	3789	3418	3177	3126	2696	2722	2364	2021
	CDH26.7	14167	861	1139	1547	1761	1609	1329	1148	1127	916	987	928	814
Wind	WSAvg	0.8	0.9	0.8	0.9	0.7	0.7	0.7	0.8	0.8	0.6	0.6	0.7	0.9
Precipitation	PrecAvg	1514	14	21	56	88	177	183	208	226	308	192	35	6
	PrecMax	1958	48	60	138	220	300	326	299	389	451	388	127	30
	PrecMin	1184	0	2	7	30	78	84	111	117	159	84	1	0
	PrecStd	183	13	16	36	50	62	52	53	60	69	68	31	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	35.2	36.5	37.8	36.2	35.4	34.2	33.6	33.8	34.2	34.1	34.4
		MCWB	23.5	24.4	25.9	26.4	26.6	26.2	25.6	26.0	26.7	26.3	25.0	24.3
	2%	DB	33.0	34.3	35.2	36.3	35.1	33.9	33.1	32.8	32.8	33.0	33.2	33.3
		MCWB	23.2	24.2	25.4	26.5	26.5	26.3	25.8	26.0	26.4	26.1	25.2	24.1
	5%	DB	32.2	33.3	34.2	35.0	34.2	32.9	32.3	32.1	31.9	32.2	32.4	32.2
		MCWB	22.9	23.9	25.3	26.4	26.5	26.2	25.8	25.9	26.2	26.1	25.2	23.8
	10%	DB	31.1	32.3	33.2	33.7	33.0	32.1	31.4	31.3	31.1	31.3	31.4	31.0
		MCWB	22.8	23.8	25.4	26.3	26.5	26.2	25.8	25.8	26.1	25.9	25.0	23.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	26.3	27.0	27.7	28.1	27.7	27.2	27.2	27.5	27.4	27.1	26.2
		MCDB	30.8	32.1	33.3	33.9	33.0	31.5	30.8	30.9	31.5	31.7	31.5	30.7
	2%	WB	25.3	25.8	26.5	27.4	27.5	27.2	26.7	26.8	27.0	26.8	26.5	25.5
		MCDB	29.7	31.0	32.6	33.3	32.3	31.2	30.4	30.5	31.0	31.0	30.8	30.1
	5%	WB	24.8	25.4	26.2	27.0	27.2	26.9	26.4	26.5	26.7	26.5	26.1	25.0
		MCDB	28.9	30.5	32.0	32.7	31.7	30.9	30.1	30.1	30.4	30.5	30.2	29.6
	10%	WB	24.2	25.1	25.8	26.7	26.8	26.6	26.2	26.2	26.3	26.2	25.6	24.4
		MCDB	28.0	30.0	31.2	31.8	31.0	30.3	29.6	29.7	29.8	30.0	29.4	28.9
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.0	10.1	9.5	7.8	7.1	6.6	6.9	6.8	7.5	8.9	10.7
		MCDBR	12.2	12.0	11.5	10.8	9.3	8.2	7.8	7.8	8.1	8.8	10.1	11.7
		MCWBR	3.7	3.3	3.1	2.5	2.2	2.1	2.0	2.2	2.6	2.9	3.1	3.6
	5% WB	MCDBR	9.8	10.4	10.2	9.8	8.2	7.3	6.8	7.2	7.6	7.8	8.6	10.0
		MCWBR	3.5	3.1	3.1	2.7	2.6	2.4	2.4	2.5	2.7	2.8	3.0	3.5
Clear-Sky Solar Irradiance	taub	0.534	0.612	0.632	0.572	0.496	0.474	0.469	0.471	0.488	0.546	0.490	0.479	
	taud	1.962	1.795	1.788	1.971	2.233	2.293	2.309	2.289	2.230	2.032	2.172	2.154	
	Ebn at Noon	766	723	722	764	811	820	827	835	825	768	801	805	
	Edh at Noon	180	221	227	189	143	133	131	135	144	172	145	146	
All-Sky Solar Radiation	RadAvg	4.94	5.13	5.23	5.51	5.04	4.85	4.63	4.67	4.57	4.72	4.88	4.87	
	RadStd	0.31	0.25	0.29	0.29	0.28	0.20	0.23	0.28	0.23	0.28	0.30	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	+0.24	+0.25	N/A	N/A	+74	+74

Nomenclature: See separate page